

Sri Lanka Institute of Information Technology

Deep Learning – SE4050

Deep Learning Project Report

Plant Disease Detection using VGG16 Model

Jameela M.J.F | IT22060662

1. Introduction to the Problem

Plant diseases are a major threat to global food security, causing significant crop losses annually. Early and accurate detection of these diseases is crucial for effective management and prevention of widespread damage. This project addresses the challenge of automated plant disease classification by leveraging the power of deep learning. Specifically, we utilize the PlantVillage dataset to train and evaluate a VGG16-based deep learning model for identifying various plant diseases. The objective is to develop an accurate and efficient model that can automatically classify plant diseases from leaf images, providing a valuable tool for farmers and agricultural professionals.

2. Background Information about the Algorithm

VGG16 is a Convolutional Neural Network (CNN) architecture proposed by the Visual Geometry Group at Oxford University. It is known for its simplicity and uniform architecture using 3x3 convolutional filters stacked on top of each other in increasing depth. The model consists of 16 weight layers — 13 convolutional layers and 3 fully connected layers — followed by a softmax layer for classification. Its key strengths include uniform design, deeper feature extraction, and transfer learning capability through pretrained weights on ImageNet.

Key Components:

- Convolutional Layers – Extract hierarchical visual features.
- Pooling Layers – Reduce dimensionality and preserve essential patterns.
- Fully Connected Layers – Perform high-level reasoning.
- Softmax Layer – Outputs class probabilities.

3. Detailed Analysis of the Dataset

The dataset used for this project is the **PlantVillage dataset**, a large collection of over 54,000 labeled images of healthy and diseased plant leaves across 38 different classes. The dataset includes species such as Apple, Tomato, Grape, and Potato with various diseases. Each image was resized and normalized to suit the VGG16 model input format.

Sample Preprocessing Steps:

- Resized images to (224, 224, 3)
- Normalized pixel values to [0, 1]
- Split into training (80%), validation (10%), and testing (10%) sets
- Applied data augmentation (rotation, flipping, zooming) to improve generalization

4. Feature Selection and Pre-processing Techniques

VGG16 relies on raw pixel intensities as features. Preprocessing ensures that the input data is clean and standardized. In this project, we applied image resizing, normalization, and data augmentation. The Keras ImageDataGenerator was used to generate batches of augmented image

data.

Code Snippet:

```
train_datagen = ImageDataGenerator(rescale=1./255, rotation_range=20,
width_shift_range=0.2, height_shift_range=0.2,
horizontal_flip=True, validation_split=0.2)
```

5. Model Architecture

The VGG16 model was implemented using TensorFlow Keras. We used transfer learning by importing the pretrained VGG16 model (excluding its top layers) and adding a custom classifier at the end for our dataset. This approach reduces training time and improves accuracy.

Code Snippet:

```
base_model = VGG16(weights='imagenet', include_top=False,
input_shape=(224,224,3))
for layer in base_model.layers:
    layer.trainable = False
x = base_model.output
x = Flatten()(x)
x = Dense(256, activation='relu')(x)
x = Dropout(0.5)(x)
predictions = Dense(38, activation='softmax')(x)
model = Model(inputs=base_model.input, outputs=predictions)
```

The model was compiled using the Adam optimizer and categorical cross-entropy loss function.

6. Model Training

The model was trained for 25 epochs with a batch size of 32. Early stopping and model checkpointing were used to prevent overfitting. During training, accuracy and loss were monitored for both training and validation sets.

```
model.compile(optimizer='adam', loss='categorical_crossentropy',
metrics=['accuracy'])
history = model.fit(train_generator, validation_data=val_generator,
epochs=25)
```

7. Results and Comparison

After 25 epochs, the VGG16 model achieved:

- Training Accuracy: **97.2%**

- Validation Accuracy: **94.8%**
- Test Accuracy: **94.1%**

The accuracy and loss curves indicate a steady improvement with minimal overfitting. Compared to the baseline CNN (which achieved 90.85% test accuracy), the VGG16 model demonstrates superior generalization and faster convergence due to transfer learning.

Visualization (Accuracy Trend):

Training and validation accuracy curves show consistent growth, indicating the effectiveness of feature reuse from pretrained layers.

8. Conclusion

The VGG16-based model effectively classified plant diseases from leaf images with high accuracy, demonstrating the power of transfer learning in agricultural applications. This approach significantly reduces the need for large datasets and long training times compared to training CNNs from scratch. The system provides a promising foundation for real-world plant disease diagnosis tools, contributing to improved agricultural efficiency and sustainability.